COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 7 : SLO – 2 | BCD addition
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I. Match the Following

• 1. BCD — b) Binary representation of decimal digits
• 2. Correction rule — a) Add 0110 (6) if sum > 9
• 3. 1001 (BCD) + 0001 (BCD) — c) 1010 (Invalid BCD)
• 4. Valid BCD digit range — d) 0000 to 1001
• 5. Decimal 25 in BCD — e) 0010 0101

II. Fill in the Blanks

• 1. Binary Coded Decimal 2. 4 3. 0000 to 1001 (0–9)
• 4. add 0110 (6) 5. 0001 0000 (i.e. decimal 10: 1010 → +0110 → 0000 with carry 1)

III. Activity Question — BCD addition of 0101 (5) and 0111 (7)

• Steps of BCD addition: (i) add the two 4-bit codes as plain binary; (ii) if the sum exceeds 1001 (9) or produces a carry, add the correction factor 0110; (iii) any carry out becomes the next higher BCD digit (0001).
• Perform: 0101 + 0111 = 1100 (12 — invalid BCD). Correction: 1100 + 0110 = 1 0010.
• Result: carry 1 to the tens digit and 0010 in the units digit → 0001 0010 = 12 in BCD. (5 + 7 = 12 ✓)